

Study III Design Pack

Coverage, power and a frozen confirmation specification

Does current Asian realised variance add information about London realised variance beyond HAR?

XAUUSD Research project | 5 October 2026
Study II permanently frozen; 2024+ holdout unopened.

Completed design work. Confirmation is not yet feasible.

The audit verifies 2,922 OANDA source files and 2,789,384 complete paired M1 candles. Thirty-two randomly selected gap/control hours were rechecked using 192 successful provider requests. No sampled missing M1 minute reappeared.

The registered conservative planning scenario needs **32,001 scored forecasts** for a 5% relative QLIKE-loss reduction at a 90% planning power target. Available pre-2024 unique weekday capacity is far smaller. These are pilot-based assumptions, not a universal minimum sample size.

The confirmation specification is frozen, but its capacity condition fails. No new confirmation forecasts or trading translation have been run. No profitable strategy or 90% joint-gate power is claimed.

Research and implementation assistance: Codex. Full audit ledgers, sample draws, code, source manifests and power sensitivities accompany this pack.

1. Why early-year eligibility is poor

Year	Weekday M1 received	Asia M1 received	Asia complete M5	Asia valid dates
2016	91.3%	96.5%	89.4%	91 / 261
2017	90.8%	94.3%	82.9%	30 / 260
2018	91.8%	96.7%	90.5%	79 / 261
2019	88.5%	91.2%	73.9%	37 / 261
2020	92.2%	97.6%	94.6%	173 / 262
2021	92.2%	98.2%	96.1%	212 / 261
2022	92.9%	98.9%	98.0%	235 / 260
2023	92.7%	98.8%	98.6%	256 / 260

The broad received-minute rate does not collapse: across UTC weekdays it ranges from 88.5% to 92.9%. The severe change is in Asian-session completeness and the resulting eligibility under fixed rules.

In 2017 Asia receives 94.3% of wall-clock M1 slots and only 82.9% of complete M5 blocks; the thresholds are 98% and 95%. In 2019 those rates are 91.2% and 73.9%. Even a scattered missing minute invalidates its entire M5 block. In 2023 the corresponding rates are 98.8% and 98.6%.

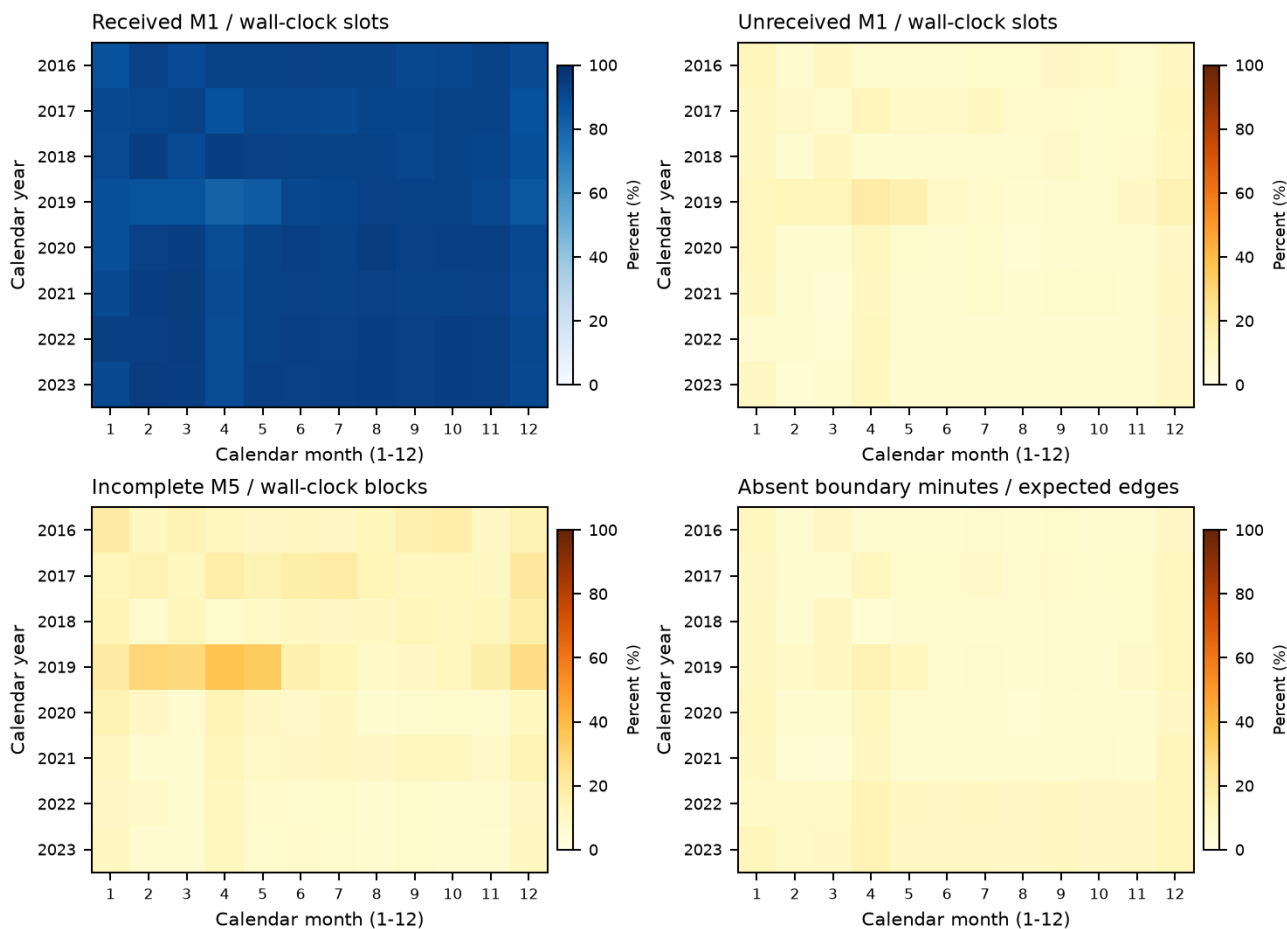
Study II first-failure labels show 2017 Asia losing 201 dates to insufficient M1 and another 29 to incomplete M5. For 2019 the corresponding losses are 199 and 25. Independent availability diagnostics also count missing boundaries; a first-failure reason alone cannot describe all defects on a date.

The calendar grid is a denominator for availability, not a trading-hours calendar. UTC weekdays include maintenance and holidays; weekends are shown separately. Unreceived minutes must not be labelled outages automatically.

The 2017 report count is 30 forecast-eligible dates, 25 with valid London outcomes. Across 2016-2021: 580 eligible feature dates, 560 labelled dates, 400 initial training observations and 160 scored forecasts. Those counts remain unchanged.

2. Monthly availability

OANDA weekday availability by calendar year and month



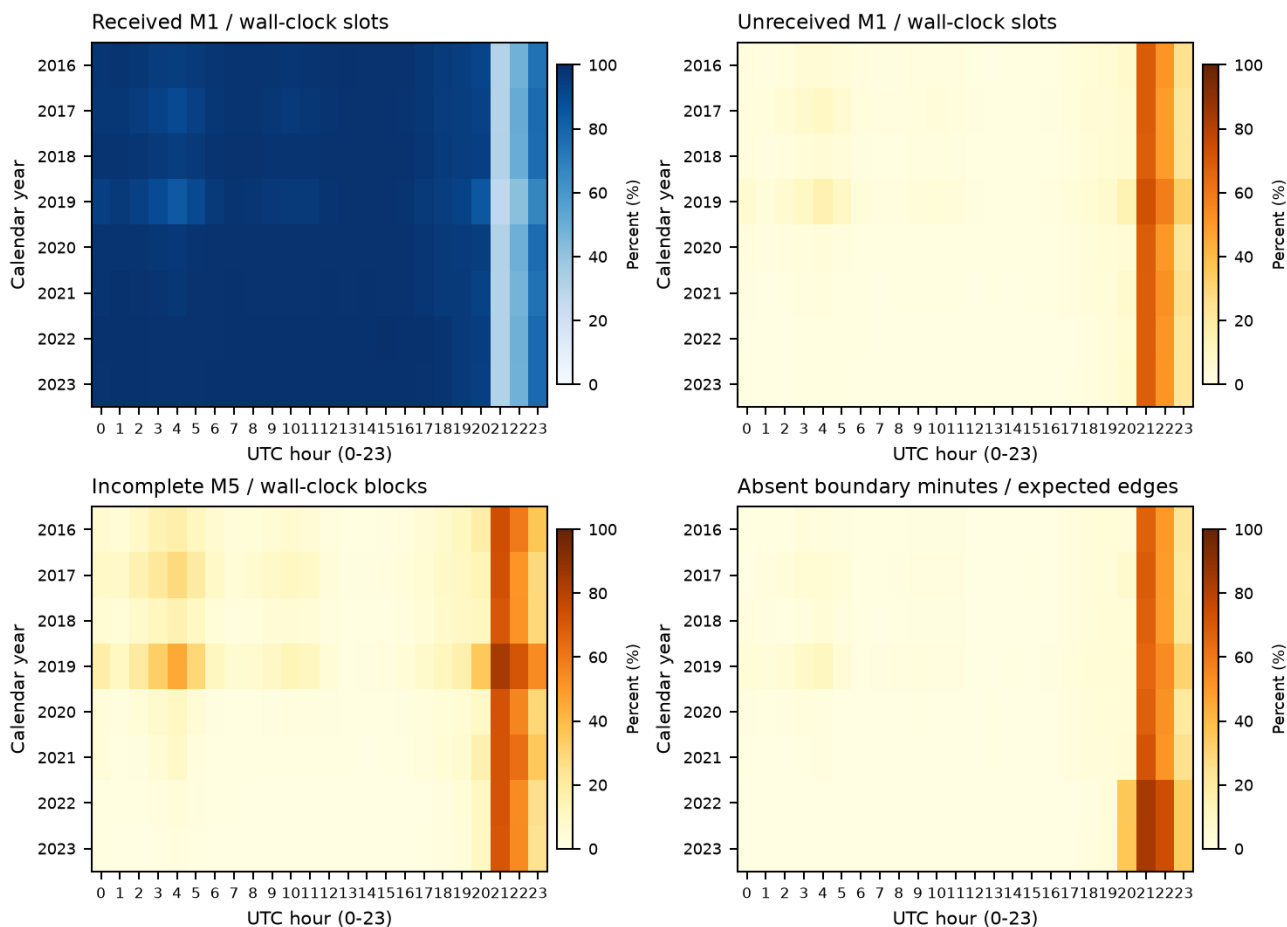
Source: immutable OANDA paired M1 responses, 2016-2023; UTC weekdays, including holiday/maintenance periods. Ratios pool all UTC hours. CSV tables include numerator and denominator counts.

Four views use the same grid: received M1, unreceived M1, incomplete M5 blocks and absent UTC-hour edge minutes. The seasonal/date structure belongs to the received provider history; it is not inferred from a backtest outcome.

Boundary minutes here mean minute 00 and minute 59 of each UTC hour. Separate daily ledgers explicitly test Asia 00:00/05:59 and London 08:00/11:59 in Europe/London, with historical DST.

3. Where in the UTC day gaps occur

OANDA weekday availability by calendar year and UTC hour

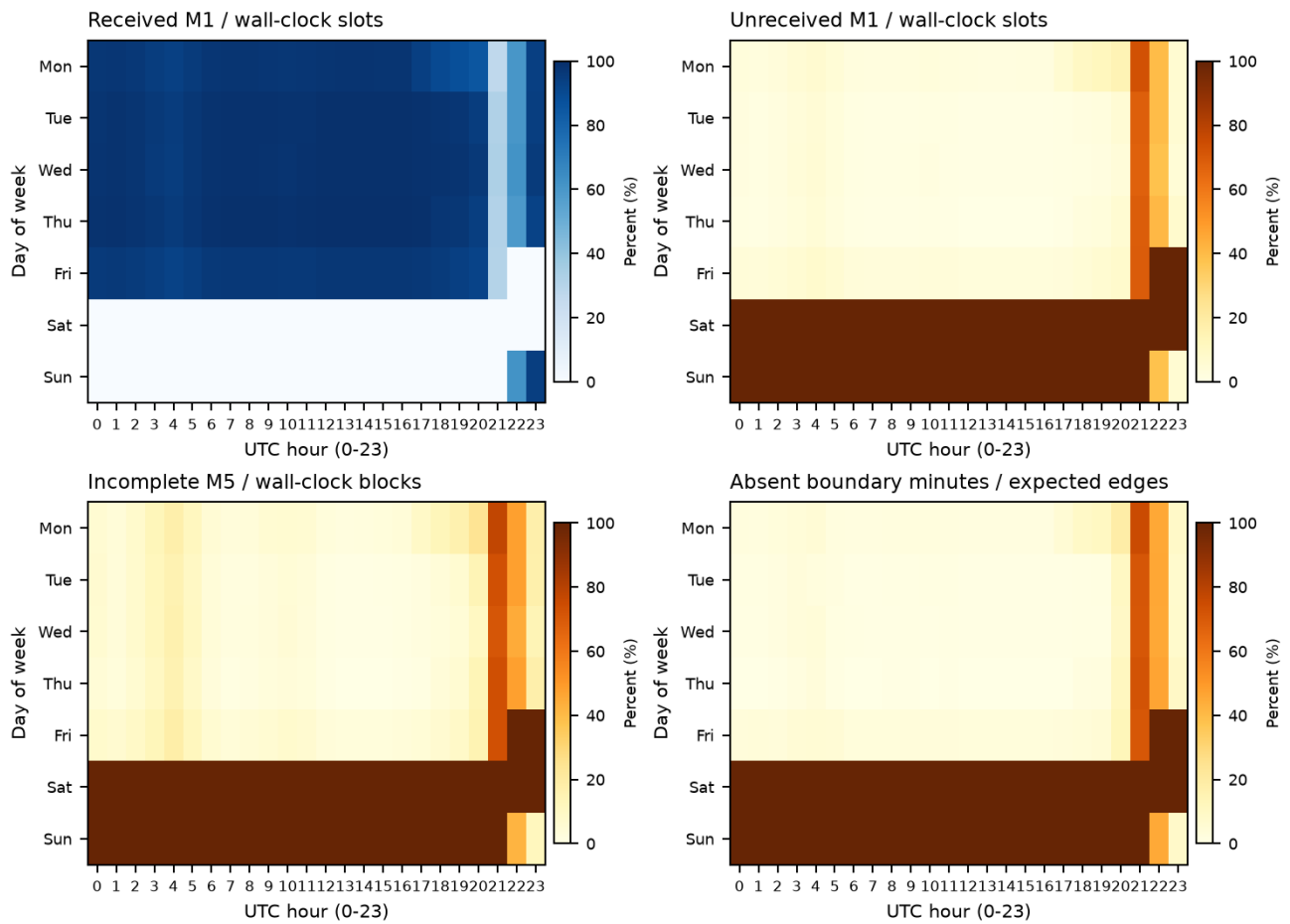


Source: OANDA 2016-2023 UTC weekdays. Asia is UTC00-06; London starts at UTC07 or UTC08 according to Europe/London DST. M5 completeness requires five received M1 candles.

Early Asian hours are less complete than active daytime hours, with stronger missing-minute penalties after M5 aggregation. Concentrated missing hours can also reflect scheduled closures; this audit does not assign an outage cause from the heatmap alone.

4. Weekday and weekend structure

OANDA availability by day of week and UTC hour



Source: all 2,922 OANDA UTC calendar dates, 2016-2023. Saturday/Sunday are included to expose market-closure structure; missing wall-clock slots are not automatically erroneous data.

Weekend and edge-of-week absences dominate some hours. Published tables provide both all-calendar and weekday-only versions by year, year/month, day of week and UTC hour. The eligibility policy still uses research weekdays and the frozen session windows.

5. Random provider rechecks

Diagnostic	Observed result
Random sample	24 missing hours + 8 complete controls; seed 20261005
Request variants	Whole-day M1 BA; hourly M1 BA/B/A/M; native hourly M5 BA
Successful queries	192 / 192 HTTP 200
Sampled missing M1 minutes	259
Recovered missing M1 minutes	0
Repeated/split BA and side sets	All 24 gap hours have identical received-minute sets
Shared price changes	0
Complete controls	All 8 repeat identically
Native M5 versus complete-M1 aggregation	250 native M5 vs 202 complete reconstructed M5; 20/24 hours supply all 12 native M5 candles

The sample was frozen before requery. Sampling is uniform over candidate hour periods within year/session strata, not weighted over missing minutes. Three sampled hours have no received M1 at all and contribute 180 of the 259 missing slots. This is a diagnostic sample, not an estimate of all historical gap incidence.

The same gaps recur across full-day, narrower-hour and separate-side requests. The sampled gaps therefore do not look like an obvious failed bulk-download or BA-combination error. Their deeper cause remains unresolved: upstream omissions, no quote updates, retained-history policy or market closure need provider/tick evidence.

Native M5 often exists where an M1-complete reconstruction fails. This supports investigating interval construction, but native M5 is not substituted into frozen Study II or Study III targets. Its return convention and economic comparability require a separately registered design.

6. Frozen power and precision method

Pilot window	N	Mean HAR QLIKE	Relative observed gain	HAC20 LRV
validation	479	0.1420	20.0%	1.584
development_oos	160	0.1214	20.6%	0.640
validation_2022	228	0.1313	10.3%	2.000
validation_2023	251	0.1518	27.5%	0.946

Already viewed Study II supplies nuisance calibration only. The primary series is HAR loss minus HAR+Asia loss, divided by that pilot window's mean HAR QLIKE. Its observed mean is removed before resampling. No pilot observation counts as new confirmation evidence.

Primary planning alternative: 5% relative QLIKE-loss reduction; sensitivity alternatives: 1%, 3%, 8% and 10%. Power targets: 80% and 90%; primary 90%. One-sided alpha is 0.025. The 5% target is operationally material for this research, but has no established conversion to trading dollars.

Estimate long-run variance with Bartlett-HAC lags 0/5/10/20/60 and circular-block-sum variance at blocks 5/10/20. For each pilot window/block pair, draw 2,000 circular resamples and take the 95th percentile of max(HAC5,HAC10,HAC20).

The largest upper estimate is 5.076, from validation_2022, block 20. The prespecified 1.5 variance-transport stress gives planning $\Omega = 7.614$. It is a conservative design assumption, not a measured provider effect.

$N = \text{ceil}[(z(0.975) + z(\text{power}))^2 \times \Omega / \text{effect}^2]$
This is scored forecast count, excluding training. Normal 95% relative-effect half-width is $1.96 \times \sqrt{\Omega/N}$.

Planning source/config were committed before the calculation. Pilot selection, serial nonstationarity and nested-model estimation limit transport of this normal approximation. Power applies to the positive-mean criterion, not the intersection of every acceptance gate.

7. Sample requirements and sensitivity

Assumed relative effect	80% planning power	90% planning power
1%	597,604	800,021
3%	66,401	88,892
5%	23,905	32,001
8%	9,338	12,501
10%	5,977	8,001

Primary requirement: **32,001** scored forecasts. At that N, normal 95% half-width is 3.02 percentage points on the fixed pilot-relative scale.

Less conservative validation-only assumption	Required N at 5% / 90%
iid 0	5,510
Bartlett-HAC 5	5,781
Bartlett-HAC 10	6,431
Bartlett-HAC 20	6,659
Bartlett-HAC 60	8,549

The primary requirement is deliberately larger than validation-only point estimates because it includes nuisance-estimation uncertainty, worse pilot-year noise and the frozen transport stress. The sensitivity table demonstrates that N depends strongly on effect and dependence assumptions; 32,001 is not a finance constant.

At 250 usable forecasts per year, the primary requirement corresponds to about 128 years; 200/year requires about 160 years. These are capacity scenarios, not an assertion that such history exists. Every scored date is one session outcome; multiple brokers on the same dates do not create independent market histories.

8. Calibration and acquisition feasibility

Target / N	Block	Conditional simulated power	MC standard error
80% / 23,905	5	97.97%	0.100 pp
80% / 23,905	10	97.27%	0.115 pp
80% / 23,905	20	96.99%	0.121 pp
90% / 32,001	5	99.91%	0.021 pp
90% / 32,001	10	99.86%	0.027 pp
90% / 32,001	20	99.83%	0.029 pp

Each row uses 20,000 centred pilot circular-block draws plus an assumed 5% location shift. Rejection uses the conservative known planning variance. Higher simulated rejection rates reflect the smaller noise of the pooled validation pilot relative to the stressed design variance.

These are simulated forecast-loss planning samples, not invented market prices or trading profits. They do not refit the nested models, establish real future power or estimate the probability that all acceptance gates pass.

Capacity upper bound	Unique scored dates
Primary untouched-time window: 1999-2015	At most 4,035 after 400 training dates
Even all 1999-2023, including previously viewed years	At most 6,121 after training
Frozen conservative requirement	32,001

These are generous weekday-calendar ceilings at 100% coverage, not authenticated provider-history claims. They already fall below the requirement. A better pre-2024 feed alone cannot make this particular powered design feasible.

The capacity gate therefore stops bulk acquisition and confirmation evaluation. No threshold or effect assumption is lowered to turn the design green. A different precision objective or prospective design would require a separate, explicit registration; the holdout is not a capacity remedy.

9. Frozen HAR confirmation specification

Item	Registered specification
Primary / secondary	HAR vs HAR+Asia primary; basic vs basic+Asia secondary
Target	Unannualised London midpoint M5 realised variance
HAR regressors	Log prior London variance, prior UTC-day variance, prior 5- and 20-session mean London variance
Added feature	Log current Asian realised variance; no directional or regime search
Clocks	Tokyo 09:00-15:00; London 08:00-12:00 Europe/London, historical DST
Fresh time / source	New independent provider, actual available dates within 1999-2015
Training / confirmation	First 400 eligible labelled dates train once; freeze coefficients/smearing; score subsequent dates
Secondary provider replication	2016-2023 on new source, reported separately; not pooled into fresh-period N
Quality rules	Study II M1/M5 completeness, gap, boundary and quote flags unchanged; no inserted prices

Acceptance requires the power-derived unique sample count; direct independent source provenance; at least 95% qualifying outcomes among issued forecasts; positive frozen-training Asia coefficient; positive mean primary QLIKE advantage with lower 95% paired block bounds above zero for blocks 5/10/20; positive advantage in each full scored year; and no more than 5% MSE/MAE deterioration.

Forecasts are issued from origin-time eligibility regardless of future label quality. Preserve missing positions in blocks and disclose selection/missingness. Report annual and cumulative loss differences, relative effect intervals and both model pairs. Secondary comparisons cannot rescue a failed primary.

Inference concerns fixed forecast algorithms conditional on initial training. Conventional DM remains descriptive; Clark-West on nested linear log-MSPE is secondary and never applied mechanically to QLIKE. Passing a positive-effect gate does not prove improvement exceeds the 5% planning alternative.

10. Source plan, holdout and permanent records

Source route	Verified state
legacy_candle	HTTP 429; no price outcome evaluated
provider_json	HTTP 429; no price outcome evaluated
Dukascopy documented S3	Requester-pays route requires AWS credentials/budget; not used
JForex/provider broker export	Authenticated export is a possible route; access/history not yet verified
OANDA / archive mirror	OANDA is the pilot; an archive mirror is not independent confirmation

Priority: direct Dukascopy paired M1 or ordered ticks, then authenticated provider export. A different broker would need documented upstream independence, symbol identity, paired quotes, timestamps, coverage and a source amendment before outcome evaluation. No second authenticated independent feed has been acquired.

Two dated direct-provider access probes returned 429; no retry loop or alternative mirror is counted as provider replication. The documented S3 route is requester-pays and requires credentials. It was not used, and no paid account or broker login was created.

All 2024-2026 data stays locked. Current code rejects windows crossing 1 January 2024. New York is secondary context, with no entry signal. C++ and MT5 EA implementation remain deferred.

The exact Study II trading candidate is carried unchanged in confirmation.json: Asian RV strictly above the trailing 60-session 80th percentile, first London Asian-boundary break 08:00-10:00, opposite-edge stop, noon exit, at most one trade. All original cost, sizing, tick and acceptance rules remain. It runs once only after confirmation passes; it is currently unrun.

If the rule eventually fails, retain a passing volatility forecast as research. Volatility-instrument monetisation or risk/execution use is a new registered study, not a redesigned spot backtest.

Study II permanent tag: study-ii-permanent-freeze-2026-10-05. Source commit: 214b50ff13c5. Archived report/bundle hashes and 51 research-file checksums remain unchanged. Planning commit ebf954e precedes power calculation; confirmation registration commit 72d2b2e precedes new-provider probes.

References

OANDA candle schema: developer.oanda.com/rest-live-v20/instrument-df
 Dukascopy export guide: dukascopy.com/wiki/en/development/data-export
 Newey-West HAC: NBER technical paper 0055
 Clark-West nested tests: Kansas City Fed working paper 05-05
 Patton QLIKE/proxy comparison: Patton (2011)